

Solve each inequality. Show your work.

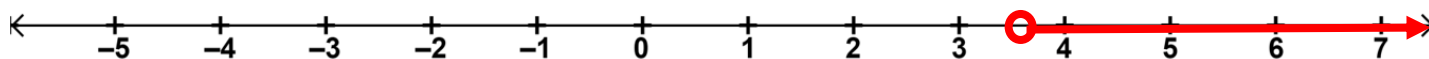
1. $-11.3 > -3x - 0.5$		2. $\frac{9}{5}x + 23 \leq -58$	
Solution		Solution	
$x > 3.6$		$x \leq -45$	
3. $\frac{2}{3}(21 - 15x) > 76$		4. $6.2(x - 4.7) \leq 31.62$	
Solution		Solution	
$x < -6.2$		$x \leq 9.8$	

Solve each inequality. Show your work. Graph the solution.

5. $9x + 7.2 > 38.7$

Solution

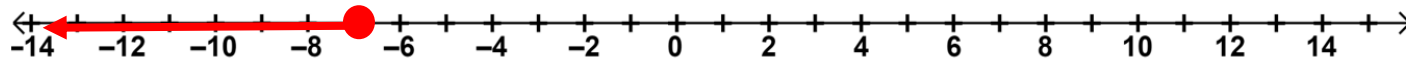
$$x > 3.5$$



6. $-8(5x + 9) \geq 208$

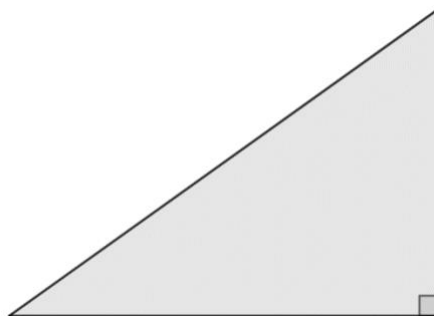
Solution

$$x \leq -7$$



7. A right triangle is shown, where the horizontal leg has a measure of 42.7 centimeters (cm) and the hypotenuse has a measure of 52.5 cm. Determine the length, in centimeters (cm), of the vertical leg. Show your work for partial credit.

Vertical Leg	30.544 cm
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8. One leg of a right triangle has a measure of 19 units. The hypotenuse has a measure of $\sqrt{7085}$ units. Determine the length, in units, of the other leg. Show your work for partial credit.

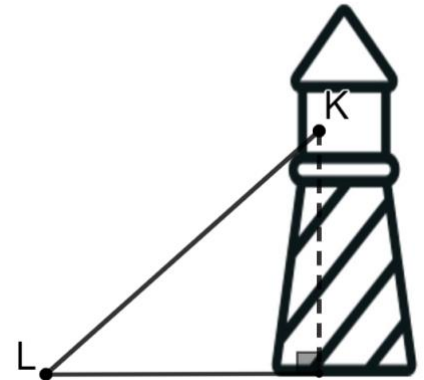
Leg	82 units
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9. The legs of a right triangle are congruent and have the measure of 23.6 yards. Determine the length, in yards, of the hypotenuse. Show your work for partial credit.

Hypotenuse	33.375 yards
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10. The Ponce Inlet Lighthouse in Ponce Inlet has a gallery deck where one can view the surrounding area. Kim is standing on the deck, point K , which is 140.75 feet (ft) above ground level. Louise, point L , is standing 200 ft from the base of the lighthouse directly below Kim's position, as shown.

Determine the distance, in feet, between point K and point L .



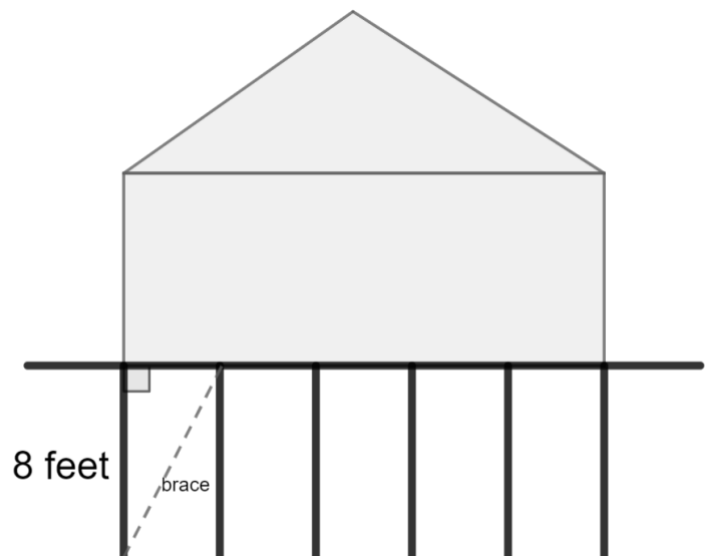
Distance between point K and point L	244.562 ft
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11. Select all of the lengths that would create a right triangle.

- ☐ 58, 156, 192
☒ 60, 91, 109
☐ 94, 168, 191
☒ 115, 252, 277
☐ 161, 240, 290

12. A house on stilts is shown where each stilt is 8 feet (ft) long. Matías built 5 braces and places each brace as shown. Each brace is 9 ft long.

Determine the length of the house, in feet.



Length of House	20.616 ft
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13. The lengths of two sides of a triangle are shown.

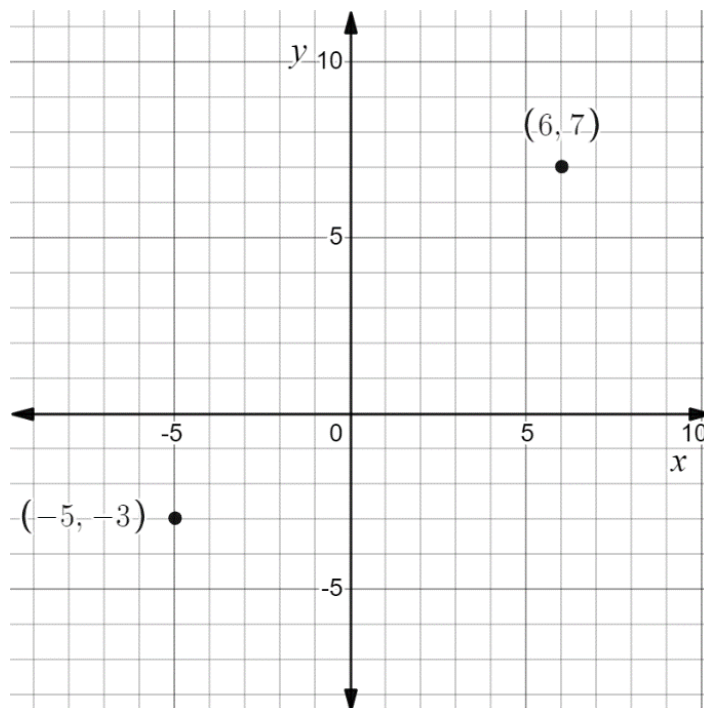
97 and 56

Complete the inequality to show the possible side measures, x , that can be used to create a triangle using the two given lengths.

Possible Side Lengths	$41 < x < 153$
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14. The points $(6,7)$ and $(-5,-3)$ are shown on the coordinate grid.

Determine the distance between the points.



Distance	$\sqrt{221}$ or 14.866
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